

Circuit Documentation

Summary

This circuit is a solar-powered system designed to manage and control various components, including a Real-Time Clock (RTC), a microcontroller (ESP32), a stepper motor system, and sensors. The circuit integrates a solar panel, a charge controller, a battery, and a buck converter to provide stable power to the components. The ESP32 microcontroller interfaces with the RTC, MPU6050 sensor, and stepper motor drivers to perform various tasks, such as timekeeping, motion sensing, and motor control.

Component List

1. **DS3231 RTC**
 - Description: A highly accurate real-time clock module with I2C interface.
 - Pins: SCL, SDA, VCC, GND
2. **BUCK Converter**
 - Description: A DC-DC step-down converter to provide stable voltage output.
 - Pins: out -, out +, IN +, in -
3. **12V 50Ah Battery**
 - Description: A rechargeable battery to store energy from the solar panel.
 - Pins: GND, 12V
4. **MPU6050**
 - Description: A 6-axis motion tracking device with accelerometer and gyroscope.
 - Pins: VCC, GND, SCL, SDA, XDA, XCL, ADO, INT
5. **ESP32 Microcontroller**
 - Description: A powerful microcontroller with Wi-Fi and Bluetooth capabilities.
 - Pins: D15, GND, 3V3, D2, D4, RX2, TX2, D5, Vin, D13, D12, D14, D27, D26, D25, D33, D32, D18, D19, D23, D22, TX0, RX0, D21, D35, D34, vn, vp, en
6. **12V 20W Solar Panel**
 - Description: A solar panel to convert sunlight into electrical energy.
 - Pins: S+, S-
7. **Bipolar Stepper Motor (NEMA 17)**
 - Description: A stepper motor used for precise control of angular position.
 - Pins: A-, A+, B+, B-
8. **A4988 Stepper Motor Driver (Red)**
 - Description: A driver module for controlling bipolar stepper motors.
 - Pins: ENA, MS1, MS2, MS3, RST, SLEEP, STEP, DIR, VMOT, GND, 2B, 2A, 1A, 1B, VCC
9. **Solar Charge Controller**
 - Description: A device to manage the charging of the battery from the solar panel.
 - Pins: Solar Cell+, Solar Cell-, Battery +, Battery -, Load+, Load-
10. **2A Fuse**
 - Description: A protective device to prevent overcurrent.
 - Pins: Terminal 1, Terminal 2

11. **L7812LC Voltage Regulator**

- Description: A voltage regulator to provide a stable 12V output.
- Pins: O/P, GND, I/P

12. **Electrolytic Capacitor**

- Description: A capacitor used for filtering and energy storage.
- Pins: -, +
- Properties: Capacitance: 0.0001 Farads

13. **Current Sensor**

- Description: A sensor to measure current flow in the circuit.
- Pins: gnd, vcc, out

Wiring Details

DS3231 RTC

- **VCC** connected to **BUCK out +**
- **GND** connected to **BUCK out -**
- **SDA** connected to **MPU6050 SDA** and **ESP32 D21**
- **SCL** connected to **MPU6050 SCL** and **ESP32 D22**

BUCK Converter

- **IN +** connected to **Solar Charge Controller Battery +** and **12V 50Ah Battery 12V**
- **in -** connected to **Solar Charge Controller Battery -** and **12V 50Ah Battery GND**
- **out +** connected to **DS3231 RTC VCC**, **MPU6050 VCC**, **ESP32 Vin**, and **A4988 Stepper Motor Driver VCC**
- **out -** connected to **DS3231 RTC GND**, **MPU6050 GND**, and **ESP32 GND**

12V 50Ah Battery

- **12V** connected to **Solar Charge Controller Battery +** and **BUCK IN +**
- **GND** connected to **Solar Charge Controller Battery -** and **BUCK in -**

MPU6050

- **VCC** connected to **BUCK out +**
- **GND** connected to **BUCK out -**
- **SDA** connected to **DS3231 RTC SDA** and **ESP32 D21**
- **SCL** connected to **DS3231 RTC SCL** and **ESP32 D22**

ESP32 Microcontroller

- **Vin** connected to **BUCK out +**
- **GND** connected to **BUCK out -**
- **D21** connected to **DS3231 RTC SDA** and **MPU6050 SDA**
- **D22** connected to **DS3231 RTC SCL** and **MPU6050 SCL**
- **D25** connected to **A4988 Stepper Motor Driver (Red) STEP**

- **D26** connected to **A4988 Stepper Motor Driver (Red) DIR**
- **D27** connected to **A4988 Stepper Motor Driver (Red) ENA**
- **D33** connected to **A4988 Stepper Motor Driver (Red) (1) STEP**
- **D32** connected to **A4988 Stepper Motor Driver (Red) (1) DIR**
- **D12** connected to **Current Sensor (2) out**
- **D14** connected to **Current Sensor (1) out**

12V 20W Solar Panel

- **S+** connected to **Current Sensor (1) vcc** and **Solar Charge Controller Solar Cell+**
- **S-** connected to **Current Sensor (1) gnd** and **Solar Charge Controller Solar Cell-**

Bipolar Stepper Motor (NEMA 17)

- **A+** connected to **A4988 Stepper Motor Driver (Red) 1A**
- **A-** connected to **A4988 Stepper Motor Driver (Red) 2A**
- **B+** connected to **A4988 Stepper Motor Driver (Red) 1B**
- **B-** connected to **A4988 Stepper Motor Driver (Red) 2B**

A4988 Stepper Motor Driver (Red)

- **VCC** connected to **BUCK out +**
- **GND** connected to **BUCK out -**
- **STEP** connected to **ESP32 D25**
- **DIR** connected to **ESP32 D26**
- **ENA** connected to **ESP32 D27**
- **MS1** connected to **ESP32 D4**
- **MS2** connected to **ESP32 D5**
- **MS3** connected to **ESP32 D15**
- **RST** connected to **ESP32 D2**
- **SLEEP** connected to **ESP32 D18**

Solar Charge Controller

- **Solar Cell+** connected to **12v 20w Solar Panel S+** and **Current Sensor (1) vcc**
- **Solar Cell-** connected to **12v 20w Solar Panel S-** and **Current Sensor (1) gnd**
- **Battery +** connected to **12V 50Ah Battery 12V** and **BUCK IN +**
- **Battery -** connected to **12V 50Ah Battery GND** and **BUCK in -**

2A Fuse

- **Terminal 1** connected to **L7812LC O/P** and **Current Sensor (2) vcc**
- **Terminal 2** connected to **A4988 Stepper Motor Driver (Red) VMOT** and **Electrolytic Capacitor +**

L7812LC Voltage Regulator

- **I/P** connected to **BUCK IN +**

- **GND** connected to **BUCK in -**
- **O/P** connected to **2A Fuse Terminal 1**

Electrolytic Capacitor

- **+** connected to **2A Fuse Terminal 2** and **A4988 Stepper Motor Driver (Red) VMOT**
- **-** connected to **A4988 Stepper Motor Driver (Red) GND**

Current Sensor

- **vcc** connected to **2A Fuse Terminal 1**
- **gnd** connected to **BUCK out -**
- **out** connected to **ESP32 D12**

Documented Code

There is no code provided for this circuit. The ESP32 microcontroller is expected to be programmed to interface with the connected components, such as reading data from the MPU6050 sensor, controlling the stepper motors via the A4988 drivers, and managing the RTC for timekeeping.